Midterm Project

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"But if thou believe
That the primordial germs of things can stop,
And in their stopping give new motions birth,
Afar thou wanderest from the road of truth.
For since they wander through the void inane,
All the primordial germs of things must needs
Be borne along, either by weight their own,
Or haply by another's blow without."
— Lucretius, *On the Nature Of Things*, Book II

1 Introduction

The time-dependent Schrodinger equation governs the evolution of closed quantum systems through the unitary operator $\hat{U}(t, t_0) = e^{-(i/\hbar)\hat{H}(t-t_0)}$. Expanding in the eigenbasis $\{\psi_n\}$ of $\hat{H}$, the state in question evolves in the form $\Psi(t) = \sum_n c_n\, e^{-iE_n t/\hbar}\psi_n$ [1]. The question then arises — if the equation is time-reversible, how do we account for the macroscopic irreversibility given by the second law of thermodynamics? [1].

In classical statistical mechanics, we understand irreversibility as a system getting stuck in the enormous amount of microstates accessible for transition. From a quantum perspective, we can strucutre this as a relationship betwen a local system and a macroscopic bath. When a given quantum state couples to a manifold of energy levels, the localized probability disperses into the continuum of possible states in the manifold [1]. Under the wide-band approximation, the dispersal created by the system is governed by Fermi's Golden Rule (FGR) [1].

This report numerically verifies these predictions by simulating two architectures—a direct relaxation model, where the initial state has access to all of the energy states int he bath, (Model A) and a doorway state model, where one state serves as a "gate" to the bath from the initial state (Model B)—and systematically identifies where finite, discrete manifolds emulate true irreversible behavior and where the approximations fail.

2 Theoretical framework

2.1 The two-level system

The isolated two-level system provides the baseline of purely reversible dynamics [1]. For states $|\phi_a\rangle$, $|\phi_b\rangle$ with energies E_a, E_b and off-diagonal coupling V_{ab}, the exact transition probability is the Rabi formula [1]:

$$P_b(t) = \frac{4|V_{ab}|^2}{(E_a - E_b)^2 + 4|V_{ab}|^2}\,\sin^2\!\left(\frac{\Omega_R}{2}\,t\right), \quad \Omega_R = \frac{1}{\hbar}\sqrt{(E_a - E_b)^2 + 4|V_{ab}|^2}. \tag{1}$$

This is purely oscillatory, and so here is no decay, relaxation, or irreversibility built into the system.

2.2 Model A: Direct relaxation into a manifold

Model A couples a single discrete state $|1\rangle$ (energy E_1) to a given N bath states $\{|k\rangle\}$ (energies E_k), with no intra-manifold couplings [1]:

$$\hat{H} = E_1|1\rangle\langle 1| + \sum_k E_k|k\rangle\langle k| + \sum_k (V_{1k}|1\rangle\langle k| + V_{k1}|k\rangle\langle 1|)\,. \tag{2}$$

$$\hat{H} = \hat{H}_0 + \hat{V}, \tag{3}$$

$$\hat{H}_0 = E_1|1\rangle\langle 1| + \sum_k E_k|k\rangle\langle k|, \qquad \hat{V} = \sum_k (V_{1k}|1\rangle\langle k| + V_{k1}|k\rangle\langle 1|)\,. \tag{4}$$

Taking the retarded Green's operator

$$\hat{G}(z) = (z - \hat{H})^{-1}, \tag{5}$$

$$\hat{G}_0(z) = (z - \hat{H}_0)^{-1}. \tag{6}$$

Using the Dyson identity,

$$\hat{G}(z) = \hat{G}_0(z) + \hat{G}_0(z)\hat{V}\hat{G}(z), \tag{7}$$

$$G_{1,1}(z) = G^{(0)}_{1,1}(z) + \sum_k G^{(0)}_{1,1}(z)V_{1k}G_{k1}(z). \tag{8}$$

$$G_{k1}(z) = G^{(0)}_{k,k}(z)V_{k1}G_{1,1}(z). \tag{9}$$

$$G_{1,1}(z) = G^{(0)}_{1,1}(z) + \sum_k G^{(0)}_{1,1}(z)V_{1k}G^{(0)}_{k,k}(z)V_{k1}G_{1,1}(z). \tag{10}$$

$$G^{(0)}_{1,1}(z) = \frac{1}{z - E_1}, \qquad G^{(0)}_{k,k}(z) = \frac{1}{z - E_k}, \tag{11}$$

we obtain

$$G_{1,1}(z) = \frac{1}{z - E_1} + \frac{1}{z - E_1}\left(\sum_k \frac{|V_{1k}|^2}{z - E_k}\right) G_{1,1}(z). \tag{12}$$

Solving for $G_{1,1}(z)$ gives

$$G_{1,1}(z) = \frac{1}{z - E_1 - B_1(z)}, \tag{13}$$

$$B_1(z) = \sum_k \frac{|V_{1k}|^2}{z - E_k}. \tag{14}$$

For the retarded Green's function we set $z = E + i\epsilon$, giving

$$G_{1,1}(E) = \frac{1}{E + i\epsilon - E_1 - B_1(E)}, \tag{15}$$

with

$$B_1(E) = \sum_k \frac{|V_{1k}|^2}{E - E_k + i\epsilon}. \tag{16}$$

Using the Sokhotski–Plemelj theorem,

$$\frac{1}{E - E_k + i\epsilon} = \mathcal{P}\frac{1}{E - E_k} - i\pi\delta(E - E_k), \tag{17}$$

$$B_1(E) = \Lambda_1(E) - \frac{i}{2}\Gamma_1(E), \tag{18}$$

$$\Lambda_1(E) = \sum_k |V_{1k}|^2 \mathcal{P}\frac{1}{E - E_k}, \tag{19}$$

$$\Gamma_1(E) = 2\pi \sum_k |V_{1k}|^2 \delta(E - E_k). \tag{20}$$

In the continuum limit we replace the sum with the density of states $\rho_K(E)$,

$$\Gamma_1(E) = 2\pi \langle |V_{1k}|^2 \rangle \rho_K(E). \tag{21}$$

Under the wide-band approximation (WBA), ρ_K and the coupling are taken to be approximately constant near E_1, giving a constant decay width Γ_1. The survival probability of state $|1\rangle$ then follows from the Fourier transform of the Green's function and decays exponentially,

$$P_1(t) = e^{-k_1 t}, \qquad k_1 = \frac{\Gamma_1}{\hbar} = \frac{2\pi}{\hbar}\,\langle |V_{1k}|^2 \rangle\, \rho_K. \tag{22}$$

2.3 Model B: Adding an intermediate

Model B introduces an intermediate state: $|0\rangle$ couples to doorway $|1\rangle$ via V_{01}, and $|1\rangle$ alone couples to $\{|k\rangle\}$ [1, 2]:

$$\hat{H} = E_0|0\rangle\langle 0| + E_1|1\rangle\langle 1| + V_{01}|0\rangle\langle 1| + V_{10}|1\rangle\langle 0| + \sum_k E_k|k\rangle\langle k| + \sum_k \big(V_{1k}|1\rangle\langle k| + \text{h.c.}\big). \tag{23}$$

Expanding $|\psi(t)\rangle = C_0|0\rangle + C_1|1\rangle$,

$$i\hbar\dot{C}_0 = E_0\, C_0 + V_{01}\, C_1, \tag{24}$$

$$i\hbar\dot{C}_1 = E_1\, C_1 + V_{10}\, C_0 + \sum_k V_{1k}\, C_k, \tag{25}$$

$$i\hbar\dot{C}_k = E_k\, C_k + V_{k1}\, C_1. \tag{26}$$

Integrating (26) formally with $C_k(0) = 0$:

$$C_k(t) = -\frac{i}{\hbar}\, V_{k1} \int_0^t e^{-iE_k(t-t')/\hbar}\, C_1(t')\, dt'. \tag{27}$$

$$i\hbar\dot{C}_1 = E_1\, C_1 + V_{10}\, C_0 - \frac{i}{\hbar}\sum_k |V_{1k}|^2 \int_0^t e^{-iE_k(t-t')/\hbar} C_1(t')\, dt'. \tag{28}$$

$$\sum_k |V_{1k}|^2 \int_0^\infty e^{-i(E_k - E_1)\tau/\hbar}\, d\tau = \int dE\; \rho(E)|V_{1E}|^2 \int_0^\infty e^{-i(E-E_1)\tau/\hbar}\, d\tau \tag{29}$$

$$\int_0^\infty e^{-i(E-E_1)\tau/\hbar}\, d\tau = \pi\hbar\,\delta(E - E_1) - i\hbar\,\mathcal{P}\left(\frac{1}{E - E_1}\right) \tag{30}$$

$$\sum_k |V_{1k}|^2 \int_0^\infty e^{-i(E_k - E_1)\tau/\hbar}\, d\tau = \frac{\hbar\Gamma_1}{2} - i\,\delta E_1, \tag{31}$$

$$\Gamma_1 = \frac{2\pi}{\hbar}\,\rho(\tilde{E}_1)|V_{1k}|^2, \tag{32}$$

$$\delta E_1 = \mathcal{P}\int \frac{\rho(E)|V_{1E}|^2}{E - E_1}\, dE, \qquad \tilde{E}_1 = E_1 + \delta E_1. \tag{33}$$

Which give us the non-Hermitian subsystem [1]:

$$\dot{C}_0 = -\tfrac{i}{\hbar} E_0\, C_0 - \tfrac{i}{\hbar} V_{01}\, C_1, \tag{34}$$

$$\dot{C}_1 = -\tfrac{i}{\hbar}\big(\tilde{E}_1 - \tfrac{i}{2}\Gamma_1\big)\, C_1 - \tfrac{i}{\hbar} V_{10}\, C_0. \tag{35}$$

For the overdamped system where $V_{01} \ll \Gamma_1$) we define slowly varying amplitudes $C_j(t) = a_j(t)\, e^{-iE_0 t/\hbar}$.

$$\dot{a}_0 = -\frac{i}{\hbar}\, V_{01}\, a_1, \tag{36}$$

$$\dot{a}_1 = \frac{i\Delta}{\hbar}\, a_1 - \frac{\Gamma_1}{2\hbar}\, a_1 - \frac{i}{\hbar}\, V_{10}\, a_0. \tag{37}$$

Taking $\dot{a}_1 = 0$

$$0 = \frac{i\Delta}{\hbar}\, a_1 - \frac{\Gamma_1}{2\hbar}\, a_1 - \frac{i}{\hbar}\, V_{10}\, a_0 \implies a_1 = \frac{V_{10}}{\Delta + i\Gamma_1/2}\, a_0. \tag{38}$$

$$\dot{a}_0 = -\frac{i}{\hbar}\, V_{01} \cdot \frac{V_{10}}{\Delta + i\Gamma_1/2}\, a_0 = -\frac{i|V_{01}|^2}{\hbar(\Delta + i\Gamma_1/2)}\, a_0. \tag{39}$$

Taking $P_0 = |a_0|^2$ and $\dot{P}_0 = 2\,\mathrm{Re}(\dot{a}_0\, a_0^*)$:

$$\dot{P}_0 = -\frac{2|V_{01}|^2}{\hbar}\, \mathrm{Re}\left[\frac{i(\Delta - i\Gamma_1/2)}{\Delta^2 + (\Gamma_1/2)^2}\right] P_0 = -\frac{|V_{01}|^2\Gamma_1/\hbar}{\Delta^2 + (\Gamma_1/2)^2}\, P_0. \tag{40}$$

And so $P_0(t) = e^{-kt}$ with Lorentzian rate [1]:

$$k = \frac{|V_{01}|^2\, \Gamma_1/\hbar}{(E_0 - \tilde{E}_1)^2 + (\Gamma_1/2)^2}. \tag{41}$$

Underdamped regime ($V_{01} \gg \Gamma_1$). With $\Delta = 0$

$$\frac{d}{dt}\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \mathbf{M}\begin{pmatrix} a_0 \\ a_1 \end{pmatrix}, \qquad \mathbf{M} = \begin{pmatrix} 0 & -iV_{01}/\hbar \\ -iV_{10}/\hbar & -\Gamma_1/(2\hbar) \end{pmatrix}. \tag{42}$$

The characteristic equation $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$ gives

$$\lambda\left(\lambda + \frac{\Gamma_1}{2\hbar}\right) + \frac{|V_{01}|^2}{\hbar^2} = 0 \implies \lambda_\pm = -\frac{\Gamma_1}{4\hbar} \pm \sqrt{\frac{\Gamma_1^2}{16\hbar^2} - \frac{|V_{01}|^2}{\hbar^2}}. \tag{43}$$

$$\lambda_\pm \approx -\frac{\Gamma_1}{4\hbar} \pm \frac{i|V_{01}|}{\hbar}. \tag{44}$$

The general solution is $a_0(t) = A_+ e^{\lambda_+ t} + A_- e^{\lambda_- t}$. Applying initial conditions $a_0(0) = 1$, $\dot{a}_0(0) = -(i/\hbar)V_{01}\, a_1(0) = 0$:

$$A_+ + A_- = 1, \tag{45}$$

$$A_+\lambda_+ + A_-\lambda_- = 0, \tag{46}$$

$$A_\pm = \frac{\lambda_\mp}{\lambda_\mp - \lambda_\pm} \xrightarrow{V_{01} \gg \Gamma_1} \frac{1}{2}. \tag{47}$$

$$a_0(t) \approx e^{-\Gamma_1 t/(4\hbar)}\left(\frac{e^{i|V_{01}|t/\hbar} + e^{-i|V_{01}|t/\hbar}}{2}\right) = e^{-\Gamma_1 t/(4\hbar)} \cos\left(\frac{|V_{01}|\, t}{\hbar}\right), \tag{48}$$

and squaring yields the damped Rabi oscillations [1]:

$$P_0(t) = |a_0(t)|^2 \approx e^{-\Gamma_1 t/(2\hbar)} \cos^2\left(\frac{V_{10}\, t}{\hbar}\right). \tag{49}$$

2.4 Poincaré recurrences in bounded manifolds

For a manifold with uniform spacing $\epsilon = \Delta E/N$, we see recurrence at $T_{\rm rec} = 2\pi\hbar/\epsilon = 2\pi\hbar\rho$ [1, 2]. The continuum approximation, therefore, is limited to $t \ll T_{\rm rec}$.

Bixon and Jortner [2] showed that in large isolated molecules, ρ is sufficient to push $T_{\rm rec}$ beyond experimental timescales, allowing us greater freedom to explore our assumptions.

3 Computational methods

3.1 Parameter selection and physical justification

Throughout, we adopt natural units with $\hbar = 1$. By the assignment guidelines, we can also specify that the manifold $\{|k\rangle\}$ should span N uniformly spaced levels over a bandwidth ΔE, and requires $\rho^{-1} \ll V_{\rm rms} \ll \Delta E$ [1].

We set $\Delta E = 10.0$. The manifold energies are taken as $E_k \in [-\Delta E/2,\ \Delta E/2]$, centered on $E_1 = 0$. This choice ensures the system state sits well within the interior of the bath spectrum, so the self-energy $B_1(E)$ evaluated at $E = E_1$ is not dominated by edge effects. If ΔE were too small relative to $V_{\rm rms}$, the coupling would push eigenstates outside the manifold bandwidth, creating localized non-decaying eigenstates at the edges that prevent complete relaxation [1]. The condition $\Delta E \gg V_{\rm rms}$ guarantees that the wide-band approximation (WBA) holds: the coupling-weighted density of states $\langle |V_{1k}|^2\rangle \rho_K(E)$ is approximately constant across the energetic region where $G_{1,1}(E)$ has significant spectral weight.

We also set the number of bath states at $N = 150$, giving level spacing $\epsilon = \Delta E/N = 0.0667$ and a density of states $\rho = N/\Delta E = 15.0$. With these parameters, the separation ratio is $T_{\rm rec}/\tau = 88.8$. We also probe $N = 30$, 75, 300, and 400 to investigate the approach to the continuum limit.

In our simulations, we set $V_{\rm rms} = 0.10$. The lower bound on $V_{\rm rms}$ is set by the requirement $V_{\rm rms} \gg \rho^{-1} = 0.0667$. If $V_{\rm rms}$ were comparable to or smaller than the level spacing, the discrete state would hybridize and create oscillations rather than smooth decay [1]. In physical terms, the perturbation must simultaneously couple $|1\rangle$ to many bath states within a spectral width Γ_1; the number of significantly coupled levels is approximately $\Gamma_1/\epsilon = 2\pi V_{\rm rms}^2\rho^2/N \approx 14$, ensuring that decay occurs across mutliple states. The upper bound $V_{\rm rms} \ll \Delta E$ ensures that the self-energy $B_1(E)$ varies slowly over the spectral window of the pole, validating the WBA. With $V_{\rm rms} = 0.10$ and $\Delta E = 10.0$, the ratio is $V_{\rm rms}/\Delta E = 0.01$.

For model B. $E_0 = E_1 = 0$. In the overdamped regime, $V_{01} = 0.20$, giving $V_{01}/\Gamma_1 = 0.21$ and an effective Lorentzian rate $k = |V_{01}|^2\Gamma_1/(\Gamma_1/2)^2 = 0.170$ ($\tau_{\rm eff} = 5.89$). This ensures that the effective decay time is well separated from $T_{\rm rec} = 94.25$, allowing roughly 16 decay times of clean exponential behavior before any recurrence. In the underdamped regime, $V_{01} = 2.0$ ($V_{01}/\Gamma_1 = 2.12$), the internal Rabi period $\pi\hbar/V_{01} \approx 1.57$ is significantly shorter than the dissipation timescale $2/\Gamma_1 \approx 2.12$.

3.2 Coupling topologies

Three configurations are investigated:

The first is equal couplings — $V_{1k} = V_{\rm rms}$ for all k. This projects the interaction onto a single collective superposition $|B\rangle = N^{-1/2}\sum_k |k\rangle$, the. The remaining $N-1$ orthogonal combinations decouple entirely. The system thus effectively reduces to a two-level problem ($|1\rangle \leftrightarrow |B\rangle$) embedded in a finite manifold, producing sine-like recurrences rather than smooth decay.

The second is a random sign — $V_{1k} = \pm V_{\rm rms}$ with random sign, giving $\langle V\rangle = 0$ and $\sqrt{\langle V^2\rangle} = V_{\rm rms}$ exactly. This breaks the collective bright-state symmetry while preserving the exact $V_{\rm rms}$ normalization that enters the FGR rate.

The third is a gaussian random distribution for V_{1k}, so $V_{1k} \sim \mathcal{N}(0, V_{\rm rms}^2)$. This most closely mimics a physical bath, where coupling matrix elements between a bright state and a dense vibrational man-

ifold are effectively random due to the chaotic nature of high-lying vibrational wavefunctions [2]. Setting $\langle V \rangle = 0$ ensures that destructive interference suppresses the pathological coherent rephasing events characteristic of ordered arrays [1].

3.3 Time evolution via exact diagonalization

The full $(N+1)$- or $(N+2)$-dimensional Hamiltonian $\mathbf{H}$ is exactly diagonalized to extract eigenvalues $\mathbf{D}$ and the unitary eigenvector matrix $\mathbf{U}$. We then can propagate the state as:

$$|\Psi(t)\rangle = \mathbf{U}\, e^{-i\mathbf{D}t}\, \mathbf{U}^\dagger\, |\Psi(0)\rangle. \tag{50}$$

The computational cost is dominated by the $O(N^3)$ diagonalization, which is negligible for $N \leq 500$.

4 Results

4.1 Model A: Coupling topology and exponential decay

Figure 1 compares $P_1(t)$ for the three coupling topologies against the FGR prediction $e^{-\Gamma_1 t}$.

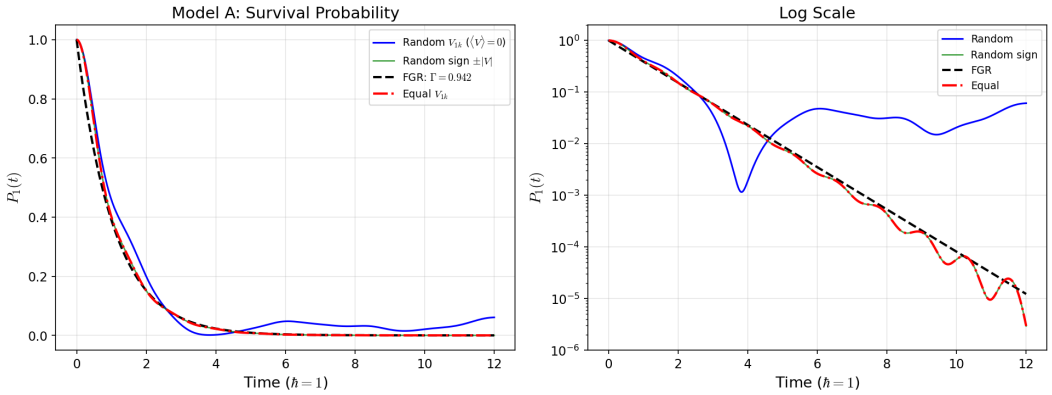

Figure 1: Model A survival probability $P_1(t)$. Left: linear scale; right: log scale. Random couplings ($\langle V \rangle = 0$, blue and green) closely track the Fermixt exponential.

Under random couplings, $P_1(t)$, we see a smooth exponential decay, similar to that mapped out by Fermi's Golden rule across the first few decay times. The log-scale panel shows $\ln P_1(t)$ is linear, suggesting the transitions are Markovian as expected. Equal couplings on the other hand create large oscillatory deviations, a deviation which could possibly be explained because identical phases map their interactiosn onto a single phase whose collective components periodically rephase. This oscillation shows that demonstrates that any attempt to create continuous decay in finite systems requires chaotic/random coupling phases, not just a high density of states.

4.2 Sensitivity analysis

The quadratic scaling $\Gamma_1 \propto V_{\text{rms}}^2$ is confirmed across all tested values (Figure 2). When $V_{\text{rms}} \to 0.40$, the decay rate ($\Gamma_1 \approx 15$) exceeds $1/\tau_c = \Delta E = 10$, and the WBA breaks down, which causes the oscillatory repetitions seen below. Increasing N at fixed ΔE (Figure 3) systematically pushes the manifold toward the continuum limit. For $N = 30$ ($\rho = 3$), the manifold has too few states to absorb probability smoothly; by $N = 400$ ($\rho = 40$), the decay is the nearly the =same as the Fermi prediction.

4.3 Poincaré recurrences

Figure 4 confirms the Poincaré recurrence theorem. After exponential evacuation, the system remains flat until $t \approx T_{\text{rec}}$, where a spike clearly emerges. Doubling N from 150 to 300 doubles T_{rec} from 94.2 to 188.5, which verifies $T_{\text{rec}} = 2\pi\hbar\rho$ [1, 2].

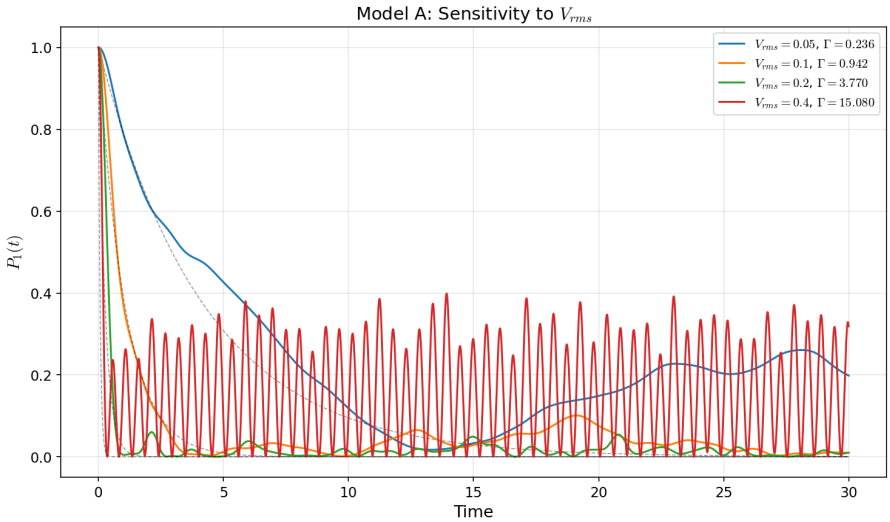

Figure 2: Model A: Dependence on V_{rms}. The FGR rate $\Gamma_1 \propto V_{\text{rms}}^2$ is confirmed across all values. At $V_{\text{rms}} = 0.40$ ($\Gamma_1 \approx 15$), the system decays faster than τ_c, breaking the WBA.

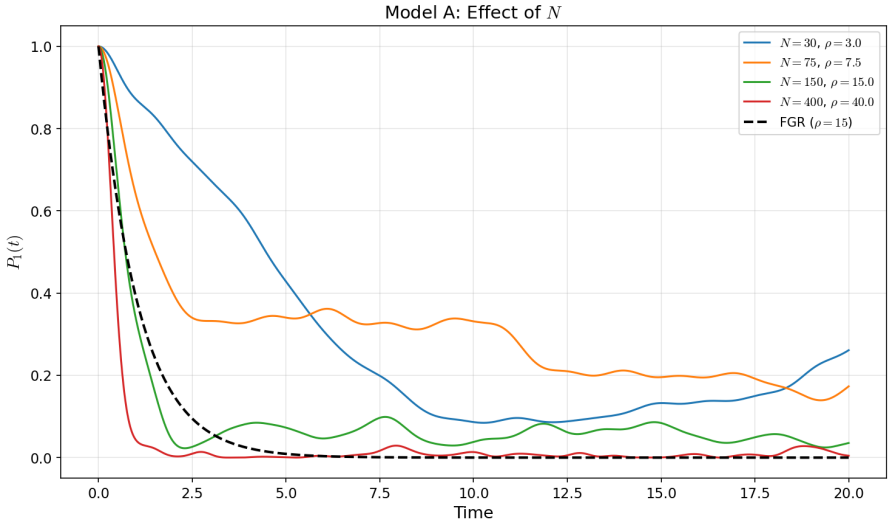

Figure 3: Model A: Effect of N at fixed $\Delta E = 10$, $V_{\text{rms}} = 0.1$. For $N = 30$ ($\rho = 3$), oscillatory deviations are conspicuous. By $N = 400$ ($\rho = 40$), the discrete manifold is indistinguishable from a true continuum.

4.4 Model B: Overdamped decay

With $V_{01} = 0.20 \ll \Gamma_1 = 0.942$ (Figure 5), $P_0(t)$ follows the analytical Lorentzian rate $k = |V_{01}|^2\Gamma_1/(\Gamma_1/2)^2 = 0.170$, yielding $\tau_{\text{eff}} = 5.89 \ll T_{\text{rec}}$. The doorway state $P_1(t)$ never exceeds $\sim$0.20, confirming the adiabatic elimination approximation—$|1\rangle$ acts purely as a transient bottleneck [1].

4.5 Model B: Underdamped Rabi oscillations

With $V_{01} = 2.0 \gg \Gamma_1 = 0.942$ (Figure 6), the population transfers in between $|0\rangle$ and $|1\rangle$ with Rabi frequency $\approx V_{01}/\hbar = 2.0$. The strong coupling hybridizes the two states into a single eigenstate before the bath reacts. Each oscillation cycle, when P_1 peaks, a fraction of the total amplitude leaks

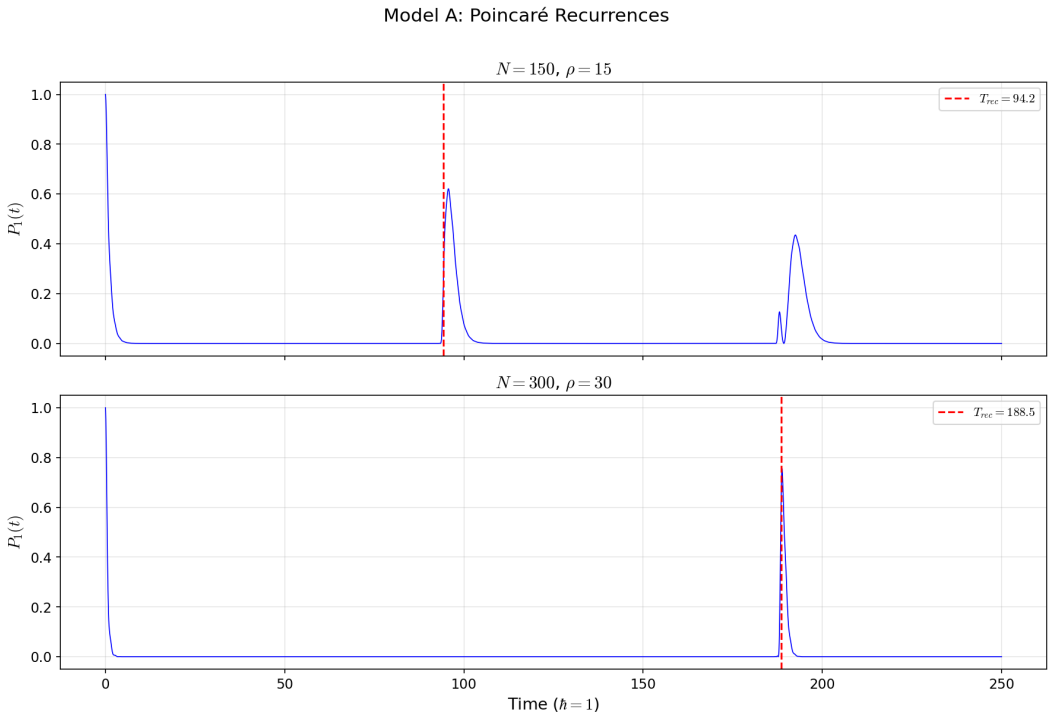

Figure 4: Model A: Long-time evolution. Top: $N = 150$, $T_{\text{rec}} = 94.2$. Bottom: $N = 300$, $T_{\text{rec}} = 188.5$. Doubling N at fixed ΔE precisely doubles T_{rec}.

into the manifold, causing the apparent stepwise decay of $P_K(t)$. The oscillation envelope decays as $e^{-\Gamma_1 t/2}$, confirming the subsystem Green's operator prediction.

4.6 Lorentzian lineshape verification

To verify Eq. (41), the effective decay was extracted across a range of detunings $E_0 - E_1$ using $N = 500$ ($\rho = 50$) and averaging over 8 random bath realizations (Figure 7). The numerical data follow the predicted Lorentzian profile, with FWHM determined by Γ_1. This confirms that the spectral width of the absorption profile is governed by the lifetime of the doorway state (lifetime broadening), which is required to fulfill the time-energy uncertainty principle [1].

4.7 Coupling topology comparison in Model B

Figure 8 confirms that the coupling topology effects observed in Model A extend to Model B: random couplings yield smooth exponential decay and monotonic manifold growth, while equal couplings introduce oscillatory artifacts and partial revivals.

4.8 Sensitivity to bandwidth ΔE

At fixed N and V_{rms}(Figure 9), reducing ΔE increases $\rho = N/\Delta E$ and therefore $\Gamma_1 = 2\pi V_{\text{rms}}^2 \rho$, while simultaneously increasing $T_{\text{rec}} = 2\pi\rho$. However, the WBA requires $\Gamma_1 \ll \Delta E$: when $\Delta E = 2$ we have $\Gamma_1 = 4.7 > \Delta E$, meaning the decay width exceeds the bath bandwidth and the self-energy $B_1(E)$ can no longer be treated as constant, which leads to the oscillations we see below.

Conversely, increasing ΔE to 30 does satisfy the WBA ($V_{\text{rms}}/\Delta E = 0.003$) but reduces ρ to 5 and T_{rec} to 31, leaving barely 10 decay times before recurrence. The right panel shows the opposite extreme: when $\Delta E = 0.5$ ($V_{\text{rms}}/\Delta E = 0.20$), the condition $\Delta E \gg V_{\text{rms}}$ is violated and population remains trapped in persistent oscillations, confirming the formation of localized edge states. The standard choice $\Delta E = 10$ thus represents the optimal balance where $V_{\text{rms}}/\Delta E = 0.01$ and $T_{\text{rec}}/\tau = 88.8$.

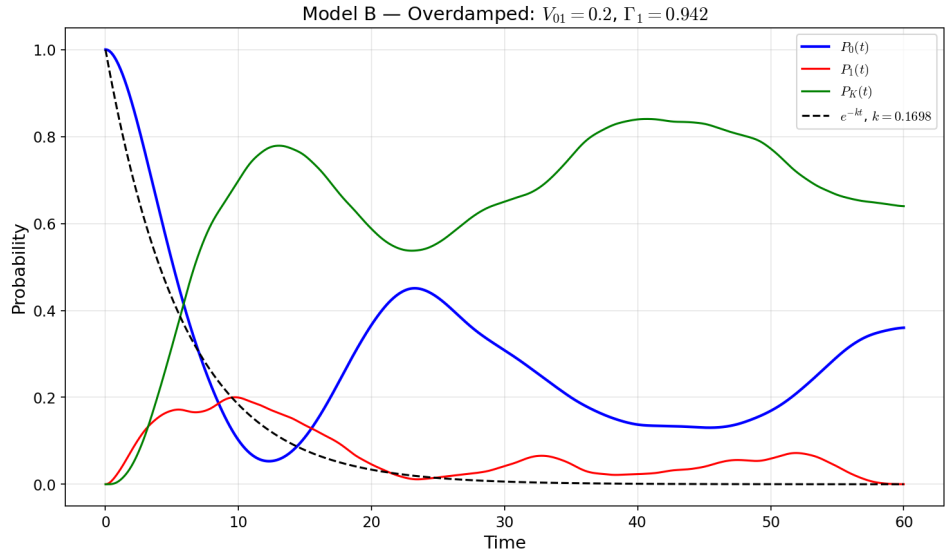

Figure 5: Model B, overdamped: $V_{01} = 0.20$, $\Gamma_1 = 0.942$ ($V_{01}/\Gamma_1 = 0.21$). $P_0(t)$ follows e^{-kt} with $k = 0.170$ ($\tau_{\text{eff}} = 5.89$). The doorway state $P_1(t)$ remains below $\sim$0.2.

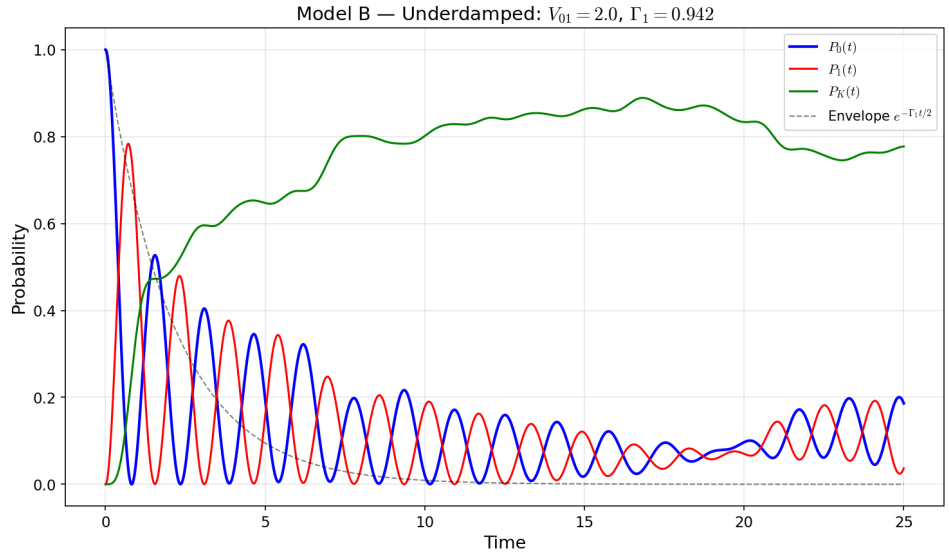

Figure 6: Model B, underdamped: $V_{01} = 2.0$, $\Gamma_1 = 0.942$ ($V_{01}/\Gamma_1 = 2.12$). High-amplitude Rabi oscillations between P_0 and P_1 decay within the envelope $e^{-\Gamma_1 t/2}$ (dashed). The manifold $P_K(t)$ grows in a characteristic staircase.

4.9 Violating the lower bound: $V_{\text{rms}} < \epsilon$

Figure 10 directly tests the lower bound of the hierarchy $V_{\text{rms}} \gg \rho^{-1} = \epsilon$. When $V_{\text{rms}} = 0.005 \ll \epsilon = 0.067$, the coupling is too weak to hybridize with more than one bath level at a time: the system undergoes small oscillations with the nearest-resonant bath state, and $P_1(t)$ remains above 0.9 over the entire simulation window.

At the crossover $V_{\text{rms}} \approx \epsilon$ (green curve), the behavior is intermediate: an initial rapid drop is followed by large revivals, reflecting coupling to a few bath states, but not enough to completely get rid of the discrete structure.

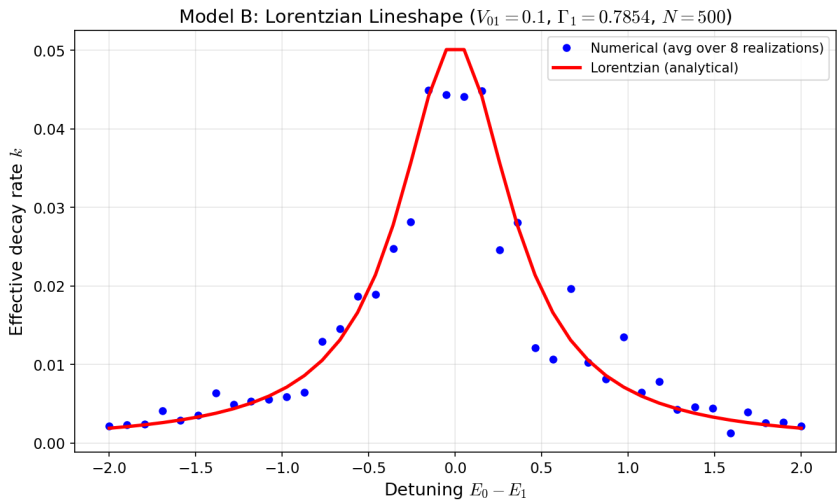

Figure 7: Model B: Effective decay rate k vs. detuning $E_0 - E_1$, compared to the analytical Lorentzian (Eq. 41). Numerical rates averaged over 8 bath realizations ($N = 500$, $V_{01} = 0.1$, $\Gamma_1 = 0.785$). FWHM is set by Γ_1.

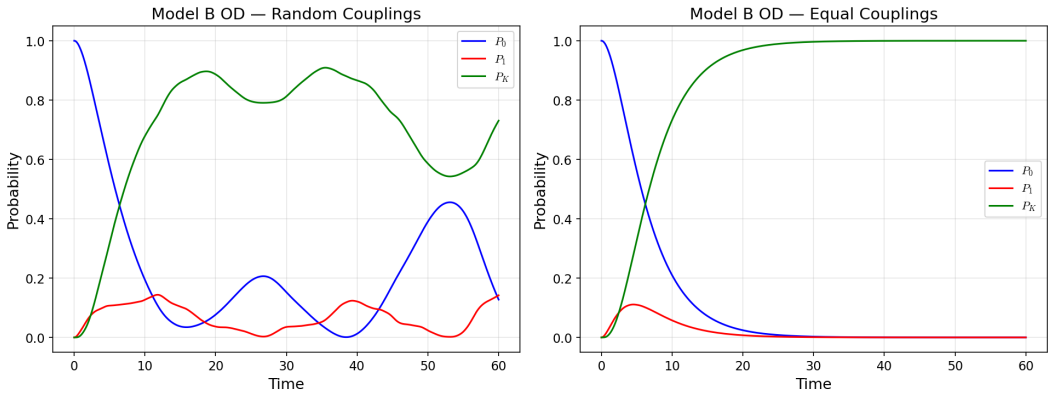

Figure 8: Model B overdamped (OD) ($V_{01} = 0.2$): random (left) vs. equal (right) bath couplings. Unlike Model A, the equal-coupling case filters the bath interaction into a single channel, while the random case shows finite-size fluctuations at a given $t > 30$.

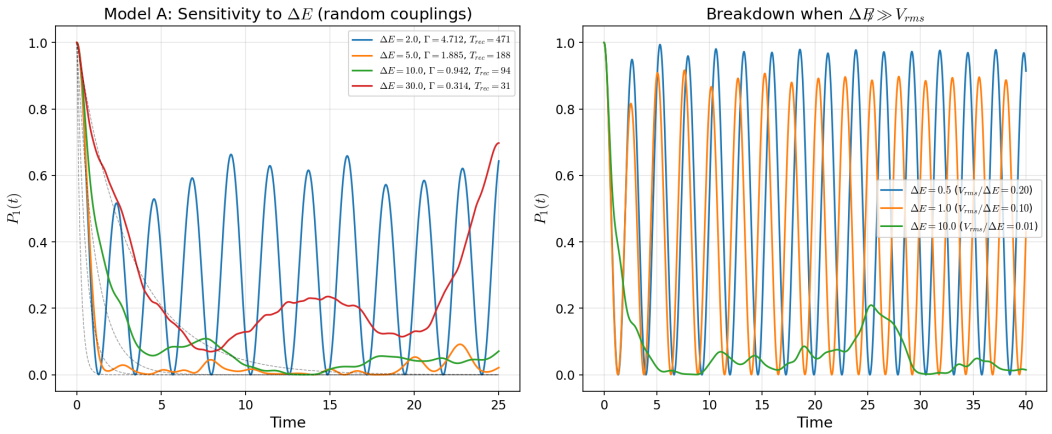

Figure 9: Model A: Sensitivity to ΔE at fixed $N = 150$, $V_{\rm rms} = 0.10$. Left: varying ΔE from 2 to 30. Reducing ΔE increases ρ and thus Γ_1, but when Γ_1 exceeds ΔE (the $\Delta E = 2$ case, where $\Gamma_1 = 4.7 > \Delta E$) the WBA breaks down to create massive oscillations. Increasing ΔE to 30 slows decay ($\Gamma_1 = 0.31$) but compresses $T_{\rm rec}$ to 31. Right: when ΔE is reduced to 0.5–1.0 ($V_{\rm rms}/\Delta E = 0.10$–$0.20$), the coupling pushes eigenstates outside the manifold bandwidth, which creates large oscillations.

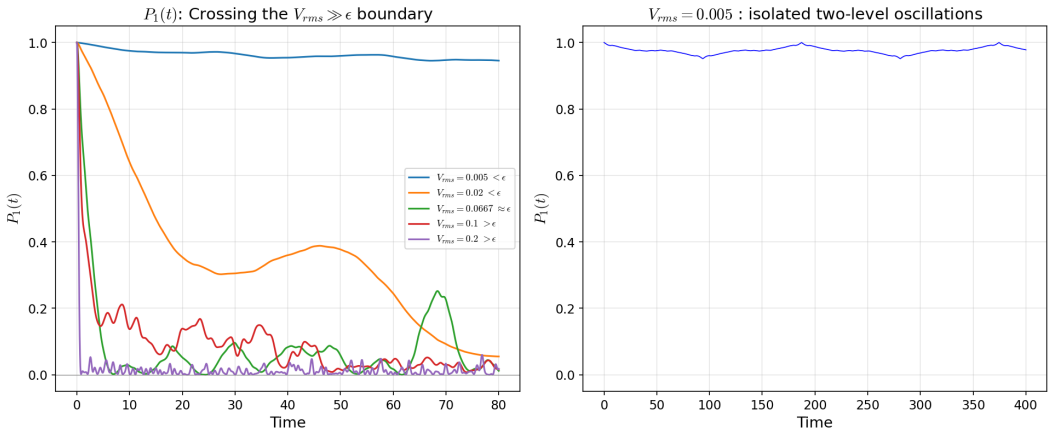

Figure 10: Model A: Behavior when $V_{\rm rms}$ crosses the $V_{\rm rms} \gg \epsilon$ boundary ($\epsilon = 0.067$). Left: transition from smooth FGR decay ($V_{\rm rms} = 0.20$, purple), to intermediate behavior ($V_{\rm rms} = 0.067 \approx \epsilon$, green) to the extreme population ($V_{\rm rms} = 0.005 \ll \epsilon$, blue). Right: zoom on $V_{\rm rms} = 0.005$, showing oscillations which suggest coupling to one or two bath states at a time.

5 Discussion and conclusion

We see that the exponential decay governed by Fermi's Golden Rule is only observed during the window $\tau_c < t \ll T_{\mathrm{rec}}$, with the additional requirements that (i) $\langle V \rangle = 0$ (random, chaotic coupling phases), and (ii) $\Delta E \gg V_{\mathrm{rms}} \gg \rho^{-1}$ [1].

A few explanations emerge for the trends we observe in the experiments above. At shorter times ($t < \tau_c$), finite bandwidth restricts the bath response speed, producing non-Markovian quadratic decay [1]. At longer times ($t \approx T_{\mathrm{rec}}$) discrete energy levels guarantee phase realignment, causing Poincaré recurrences. Doubling N precisely doubles T_{rec}, which we can see experimentally [1, 2]. Equal couplings create a single collective state whose components periodically rephase, generating periodic oscillations. Random phases prevent this from occurring [1]. When $V_{01} \gg \Gamma_1$, internal Rabi splitting dominates before relaxation can proceed, producing damped oscillations with clear staircase manifold growth [1].

These numerical experiments validate the framework based on Fermi's Golden Rule while highlighting the fragility of the continuum approximation. Fundamentally, irreversibility in finite systems is sustained only as long as the density of states pushes the inevitable quantum revivals beyond observational timescales [1, 2].

References

[1] A. Nitzan. *Chemical Dynamics in Condensed Phases: Relaxation, Transfer and Reactions in Condensed Molecular Systems*. Oxford University Press, Oxford, 2006.

[2] M. Bixon and J. Jortner. Intramolecular radiationless transitions. *J. Chem. Phys.*, 48:715–726, 1968.

A Appendix:

The complete Python simulation code (`midterm5260.py`) generates all eight figures using exact diagonalization via `numpy.linalg.eigh`. The script is included as a supplementary file.

- Bath energies are uniformly spaced: $E_k \in [-\Delta E/2,\ \Delta E/2]$.
- Time evolution is computed via $|\Psi(t)\rangle = \mathbf{U}\, e^{-i\mathbf{D}t}\, \mathbf{U}^\dagger |\Psi(0)\rangle$.
- Lorentzian lineshape data (Figure 7) uses $N = 500$, 8 random bath realizations, and log-linear rate fitting in the range $0.15 < P_0 < 0.92$.
- Random seed fixed at 42 for reproducibility.